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Ocean Spin-Up Effects on Coupled Regional Climate Simulations over CORDEX East Asia

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1. Introduction

- **East Asia** is characterized by **strong monsoon variability** and **complex coastal processes**.
- **Atmosphere-ocean interactions** are a major source of **uncertainty in regional climate simulations** over this region.
- **Sea surface temperature (SST)** and **sea surface salinity (SSS)**, influenced by **western boundary currents and marginal seas**,
  - regulate **lower-tropospheric stability**
  - modulate **moisture supply**
  - affect **precipitation** and near-surface climate variability.
- Despite their importance, **long-term coupled regional simulations** with well-constrained ocean initial conditions remain **limited**.
- **Insufficient ocean spin-up** may introduce **biases through incomplete ocean adjustment** processes.
- The role of **long ocean spin-up** in **regional coupled simulations** over **East Asia** is still **not clearly quantified**.

2. Data & Method

Model Configuration

Couple System	Coupled Ocean-Atmosphere-Wave-Sediment Transport (COAWST)
Coupler	Model Coupling Toolkit (MCT) v 2.6.0
Ocean	Regional Ocean Modeling System (ROMS) 3.9
Atmosphere	Weather Research and Forecasting Model (WRF) v 4.2.2
Wave(s)	Simulating Waves Nearshore (SWAN) v 41.31 WAVEWATCH III (WW3) v 7.12 Infragravity wave model (InWave) v 2.0
Sediment Transport	Community Sediment Transport Modeling Systems (CSTMS)

Ocean	Regional Ocean Modeling System (ROMS) v 3.9	
Horizontal Resolution | Integration Time Step	12 km (817×513) | dt = 30 s	
Ocean depth	ETOPO2	
Lateral boundary	Mixed Radiation Nudging	temperature, salinity, 3D U and V momentum (baroclinic)
	Shchepetkin	2D U and V momentum (barotropic)
	Chapman _implicit	Free-surface, every 6 hours
Boundary Layer Scheme (Vertical Turbulent Mixing)	Generic Length Scale (Umlauf and Burchard, 2003)	
Momentum and tracer diffusion	Harmonic	
River inflow / Tide	X / O	
Initial & Boundary data	GLORYS12V1 (0.083° / daily)	

Atmosphere	Weather Research and Forecasting Model (WRF) v 4.2.2	
Horizontal Resolution | Integration Time Step	12 km (825×521) | dt = 36 s	
Vertical Layers	35 hybrid levels	
Microphysics	WSM6	P3
Cumulus Parameterization	MSKF	KSAS
Radiation	RRTMG	
Planetary Boundary Layer	YSU	MYJ
Land Surface	Noah LSM	
Initial & Boundary data	ERA5 (0.25° and 6 h)	

Data

- **ERA5 (ECMWF Re-Analysis v5)** | **1-hourly** | **0.25°** | **1940 ~ near-real-time**
  Used as initial and 6-hourly lateral boundary conditions for the WRF model.
  Used as 1-hourly atmospheric surface radiative forcing for standalone ROMS initialization.
- **GLORYS (Global Ocean Physics Reanalysis)** | **daily** | **0.083°** | **1993 ~ near-real-time**
  Used as initial and 6-hourly lateral boundary conditions for the COAWST and standalone ROMS models (the 12 UTC field was also applied at 00, 06, and 18 UTC).
- **MSWEP (Multi-Source Weighted-Ensemble Precipitation)** | **3-hourly** | **0.1°** | **1979 ~ real-time**
  Accumulated to daily totals and interpolated onto the model grid for evaluation of daily accumulated precipitation.
- **RSMC (Regional Specialized Meteorological Center, Tokyo) Best Track** | **6-hourly** | **—** | **1979 ~ 2025**
  Accumulated to daily totals and interpolated onto the model grid for evaluation of daily accumulated precipitation.

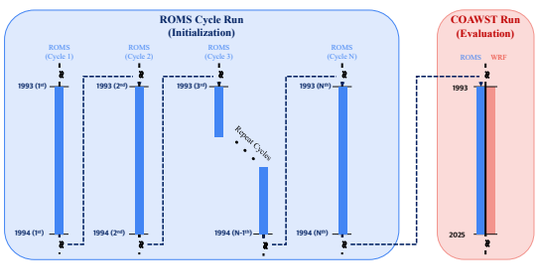
Typhoon Tracking

- The tracking criteria for model outputs follow Kim et al. (2025), with modifications considering the relatively high model resolution.
- Tracking was performed using 6-hourly model output.
- Only systems persisting for at least 2 days were considered as tropical cyclones.

Criteria	Radius Value	Threshold
Minimum SLP	500 km	Local minimum
Maximum surface wind speed		> 16 m/s
Maximum relative vorticity		> 6.0 * 10
Temperature (300, 500, 700 hPa)		Sum of deviations at 3 layers > 2.0 K
Maximum wind speed		850 hPa > 300 hPa

Ocean Spin-up

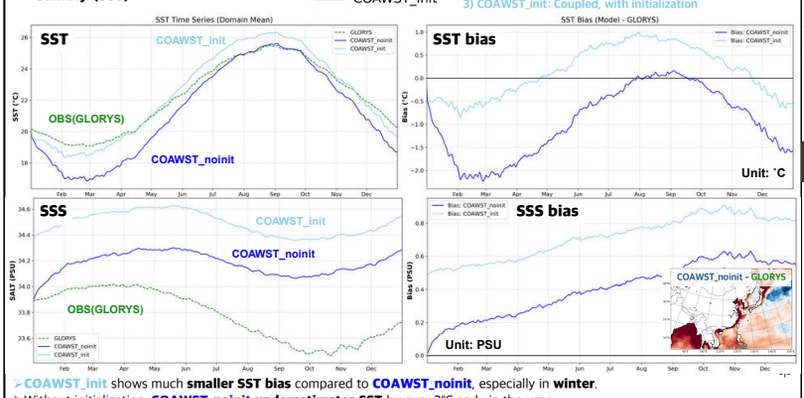
- The ocean has a much larger heat capacity compared to the atmosphere.
- It exerts a strong influence on long-term climate variability and change.
- Ocean initial conditions critically affect simulation and prediction accuracy.
- Spin-up is the process through which the model domain and intrinsic model characteristics adjust to the forcing data, reaching a statistically balanced equilibrium state.



3. Results

Sea Surface Temperature (SST), Salinity (SSS)

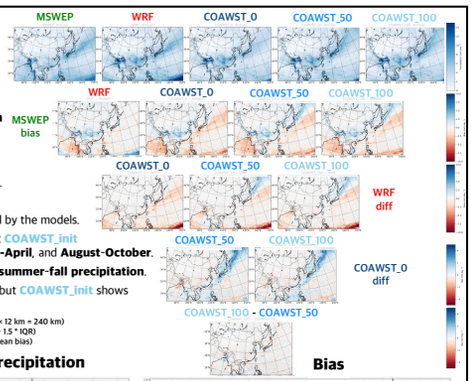
MSWEP — 0) Observation or Reanalysis (MSWEP, GLORYS, ERA5 etc.)
WRF — 1) WRF: Uncoupled, Atmosphere-only
COAWST_noinit — 2) COAWST_noinit: Coupled, without ~~initialization~~
COAWST_init — 3) COAWST_init: Coupled, with initialization



- **COAWST_init** shows much **smaller SST bias** compared to **COAWST_noinit**, especially in **winter**.
- Without initialization, **COAWST_noinit** **underestimates SST** by over 2°C early in the year.
- Initialization helps maintain **realistic SST** throughout the year, **improving model stability**.
- Although initialization was performed, the SALT bias appears to increase continuously over time.
- Seasonal variability is generally well reproduced.

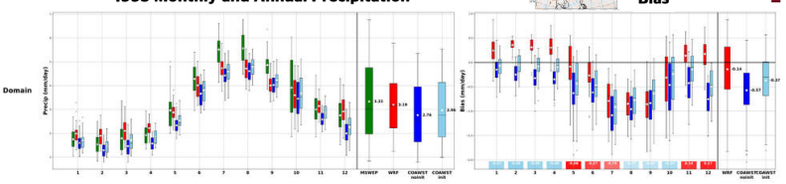
Precipitation (1993 Annual Mean)

- **All models overestimate** precipitation **over inland** regions and **underestimate** it **over coastal** areas.
- **WRF** shows **strong inland overestimation**, while **COAWST tends to underestimate over oceans**.
- **Initialization (COAWST_init) enhances overestimation** over the **northwest Pacific** and **east of Japan**.
- There are **no significant difference** between **COAWST_100** and **COAWST_50**
- Strong or weak precipitation events are not well captured — models tend to **simulate a flattened distribution**.
- **MSWEP's seasonal variability** is generally **well captured** by the models.
- **WRF** shows the **best match** in terms of **annual mean**, but **COAWST_init** performs **better** in **monthly means**, especially in **January-April**, and **August-October**.
- **WRF** tends to **overestimate winter** and **underestimate summer-fall precipitation**.
- **COAWST_noinit** generally **underestimates** precipitation, but **COAWST_init** shows **improvement**.

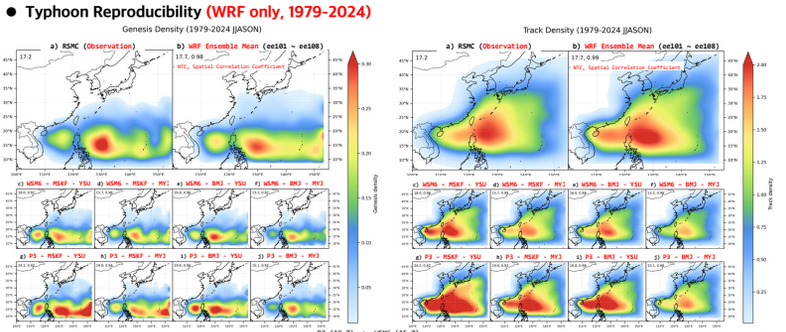


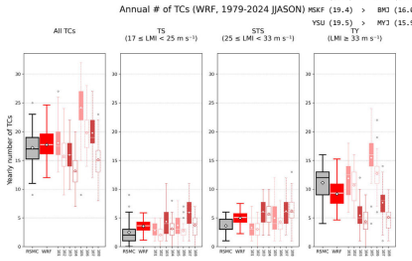
- Domain: Entire area excluding the buffer zone (20 × 12 km = 240 km)
- Box plot: (Q3 + 1.5 * IQR, Q3, median, mean, Q1, Q1 - 1.5 * IQR)
- Bottom part of bias: monthly best model (lowest mean bias)

1993 Monthly and Annual Precipitation



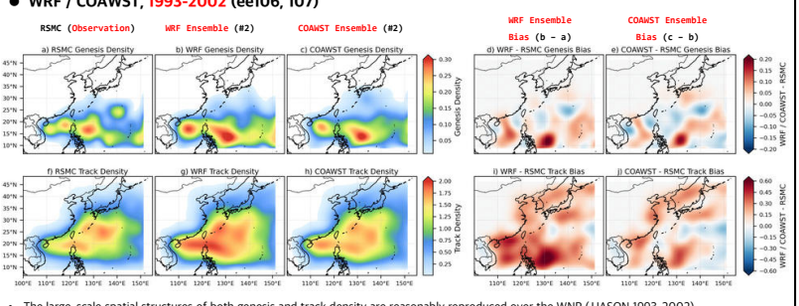
Typhoon Reproducibility (WRF only, 1979-2024)



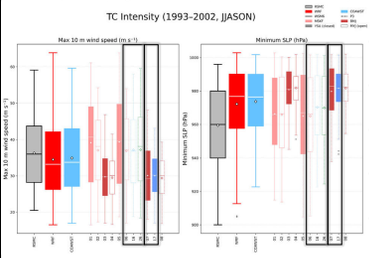


- The major tropical cyclone genesis regions and their spatial distribution are well reproduced, with a spatial correlation coefficient of 0.99 (0.98 for genesis density).
- The total number of tropical cyclones (NTC = 17.7) is slightly overestimated.
- The genesis center shows a slight eastward extension compared to observations.
- The characteristic northward-moving track pattern along the southeastern coast of Japan is realistically simulated.
- Variability exists among individual experiments, but performance improves in the ensemble mean.
- The strongest genesis density appears in ee105 and ee106 (P3-MSKF), while the weakest appears in ee104 (WSM6-BMJ-MYJ).
- The differences among physics scheme combinations in genesis density show similar tendencies in track density.

WRF / COAWST, 1993-2002 (ee106, 107)



- The large-scale spatial structures of both genesis and track density are reasonably reproduced over the WNP (JJASON 1993-2002).
- WRF exhibits widespread positive track density bias, with a noticeable northward extension of the main TC pathway.
- COAWST substantially reduces the excessive northward track density bias seen in WRF, leading to improved agreement with obs.
- Genesis density biases remain patchy and similar between both, indicating limited coupling impact on TC formation locations.
- Overall, the coupling effect is more pronounced in track density than in genesis density.



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- Coupling effect is more pronounced in track density than genesis.

4. Summary

- Ocean spin-up substantially reduces SST bias and improves model stability, while differences between 50- and 100-year spin-up are negligible. However, SALT bias gradually increases over time.
- All models overestimate inland precipitation and underestimate coastal regions. Coupled simulations with initialization improve seasonal precipitation biases relative to WRF.
- Tropical cyclone genesis and track structures are realistically reproduced (spatial correlation ≈ 0.99), although total TC frequency is slightly overestimated. COAWST reduces the excessive northward track bias seen in WRF, indicating a stronger coupling impact on track structure than on genesis.
- TC intensity remains underestimated in both models, and physics scheme sensitivity exerts a larger influence on intensity than coupling or spin-up effects.
- Given the importance of freshwater forcing in coastal and marginal seas, river discharge is expected to further influence upper-ocean stratification and regional climate responses; related sensitivity experiments are currently under analysis.

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